Discussion section 8

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1 Chp 5, Q3

Commuters in an urban center either drive by car, take the bus, or bike to work. Based on recent changes to local transportation systems, urban planners predict yearly annual behavior changes of commuters based on a Markov model with transition matrix

$$P = \begin{array}{c} \\ \text{Car} \\ \text{Bus} \\ \text{Bike} \end{array} \begin{array}{c} \begin{array}{ccc} \text{Car} & \text{Bus} & \text{Bike} \end{array} \\ \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.1 & 0.8 & 0.1 \\ 0.05 & 0.15 & 0.8 \end{pmatrix} \end{array}.$$

(a) In a city of 10000 commuters, 6000 currently drive cars, 3000 take the bus, and 1000 bike to work. Two years after transportation changes are made, how many commuters will use each type of transportation? Over the long term, how many commuters will use each type of transportation?

(b) The European Cyclists Federation reports the following levels of CO_2 emissions (in grams) per passenger per kilometer traveled: car: 271, bus: 101, bike: 21. What is the current average amount of CO_2 emissions per kilometer traveled? How does the average change over the long-term?

Proof. (a) Recall for a Markov chain $X_0, X_1, \dots$ with transition matrix P and initial distribution α, then the distribution of X_n is αP^n, i,e,

$$\mathbb{P}(X_n = j) = (\alpha P^n)_j \quad \text{for all } n.$$

We have the initial distribution $\alpha = (0.6, 0.3, 0.1)$.

The distribution in year two is $\alpha P^2 = (0.3645, 0.4165, 0.219)$.

The expected number in year two is $10000 \cdot \alpha P^2 = (3645, 4165, 2190)$.

We have the limiting(stationary) distribution $\pi = (5/24, 11/24, 1/3)$.

The expected number over the long term is $10000 \cdot \pi = (2083, 4583, 3333)$.

(b) We have the initial distribution $\alpha = (0.6, 0.3, 0.1)$. The current average of CO_2 is $0.6 \cdot 271 + 0.3 \cdot 101 + 0.1 \cdot 21 = 195$.

We have the limiting(stationary) distribution $\pi = (5/24, 11/24, 1/3)$. The long-term average of CO_2 is $\frac{5}{24} \cdot 271 + \frac{11}{24} \cdot 101 + \frac{1}{3} \cdot 21 = 109.75$.

□

2 Chp 5, Q5

Exhibit a Metropolis–Hastings algorithm to sample from the distribution

1	2	3	4	5	6
0.01	0.39	0.11	0.18	0.26	0.05

with proposal distribution based on one fair die roll.

Proof. The proposal chain has transition matrix T with $T_{ij} = \frac{1}{6}$ for any $i, j \in \{1, 2 \cdots, 6\}$.

We compute the acceptance function

$$a(i,j) = \frac{\pi_j T_{ji}}{\pi_i T_{ij}},$$

where π_i is the given probability distribution. In matrix form,

$$a(i,j) = \begin{pmatrix} 1 & 39 & 11 & 18 & 26 & 5 \\ \frac{1}{39} & 1 & \frac{11}{39} & \frac{6}{13} & \frac{2}{3} & \frac{5}{39} \\ \frac{1}{11} & \frac{39}{11} & 1 & \frac{18}{11} & \frac{26}{11} & \frac{5}{11} \\ \frac{1}{18} & \frac{13}{6} & \frac{11}{18} & 1 & \frac{13}{9} & \frac{5}{18} \\ \frac{1}{26} & \frac{3}{2} & \frac{11}{26} & \frac{9}{13} & 1 & \frac{5}{26} \\ \frac{1}{5} & \frac{39}{5} & \frac{11}{5} & \frac{18}{5} & \frac{26}{5} & 1 \end{pmatrix}.$$

By exploiting the proof of Metropolis-Hastings Algorithm in the textbook(cf. Section 5.2), the constructed Markov chain has the transition matrix, for any $i \neq j$.

$$P_{ij} = \begin{cases} T_{ij} \cdot a(i,j), \text{ if } a(i,j) \leq 1, \\ T_{ij}, \text{ if } a(i,j) > 1. \end{cases} \tag{2.1}$$

Once we know P_{ij} for $i \neq j$, then the P_{ii} terms are uniquely determined, since each row of P will add up to 1. In matrix form,

$$P = \frac{1}{6}\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ \frac{1}{39} & \frac{173}{39} & \frac{11}{39} & \frac{6}{13} & \frac{2}{3} & \frac{5}{39} \\ \frac{1}{11} & 1 & \frac{27}{11} & 1 & 1 & \frac{5}{11} \\ \frac{1}{18} & 1 & \frac{11}{18} & \frac{55}{18} & 1 & \frac{5}{18} \\ \frac{1}{26} & 1 & \frac{11}{26} & \frac{9}{13} & \frac{95}{26} & \frac{5}{26} \\ \frac{1}{5} & 1 & 1 & 1 & 1 & \frac{9}{5} \end{pmatrix}.$$

$\square$